Appendix A

Algorithm Robustness: k-modes Replication

The main analysis uses k-means clustering applied to standardised construct indices. k-means assumes Euclidean distance and continuous variables, which is a reasonable approximation for the 0–1 construct indices used here, but is not the optimal algorithm for binary data. As a robustness check, we replicate the clustering using k-modes, an algorithm designed specifically for categorical and binary data that uses Hamming distance and modal centroids rather than Euclidean distance and means.

A.1 Method

k-modes was applied directly to the 21 binary survey items (not the construct indices) using Huang initialisation and k=5 clusters to match the main solution. Ten random initialisations were run and the solution with the lowest total Hamming cost was retained. Agreement between the k-means and k-modes solutions was assessed using the Adjusted Rand Index (ARI) and Normalised Mutual Information (NMI), both corrected for chance agreement, and a cross-tabulation with optimal cluster matching via the Hungarian assignment algorithm.

A.2 Results

The k-modes solution produced five clusters of sizes 3,454 / 2,116 / 1,478 / 1,426 / 113, compared to 2,664 / 2,591 / 2,264 / 962 / 106 in the k-means solution. Overall agreement was moderate: ARI = 0.270, NMI = 0.304. Under optimal cluster matching, 55.0% of referees (4,719 / 8,587) were assigned to the same cluster by both algorithms.

Both algorithms identify the same four substantive motivational dimensions: Passion, Financial, Sport & fitness, and Family; as the defining axes of differentiation. The All-round residual is reproduced with near-perfect agreement (99.1%). The main difference is that k-modes splits the Sport & social profile into two distinct sport-oriented groups, suggesting that this is the least homogeneous cluster in the main solution.

Table A1. k-modes cluster profiles, defining constructs and mean indices.

Cluster	n	Defining	Pass.	Fin.	Car.	Soc.	Fam.	L&D	Sport	Comm.
C1(km)	3,454	Passion	**0.60**	0.13	0.25	0.42	0.05	0.09	0.29	0.03
C2(km)	2,116	Sport & fitness	0.26	0.50	0.14	0.22	0.08	0.07	**0.71**	0.07
C3(km)	1,478	Sport & fitness	0.43	0.13	0.07	0.44	0.06	0.08	**0.60**	0.06
C4(km)	1,426	Financial	0.31	**0.79**	0.21	0.22	0.06	0.08	0.16	0.03
C5(km)	113	All dimensions	0.98	0.88	0.83	0.90	0.79	0.94	0.94	0.88

Table A2. Cross-tabulation of k-means and k-modes cluster assignments

k-means cluster	n	km C1 Passion	km C2 Sport A	km C3 Sport B	km C4 Financial	km C5 All-round
C1 Sport & social	**2,664**	771 (29%)	885 (33%)	990 (37%)	17 (1%)	1 (0%)
C2 Passion-driven	**2,591**	2,182 (84%)	38 (1%)	237 (9%)	129 (5%)	5 (0%)
C3 Financial	**2,264**	192 (8%)	875 (39%)	73 (3%)	1,124 (50%)	0 (0%)
C4 Family-influenced	**962**	308 (32%)	318 (33%)	178 (18%)	156 (16%)	2 (0%)
C5 All-round	**106**	1 (1%)	0 (0%)	0 (0%)	0 (0%)	105 (99%)

Rows = k-means clusters (main solution). Columns = k-modes clusters (robustness check). Cell values show n and row percentage.

A.3 Interpretation

The Financial cluster shows the strongest convergence across algorithms (49.6% of k-means Financial members assigned to the k-modes Financial cluster, plus an additional 38.6% in a sport-adjacent k-modes cluster), as does the All-round cluster (99.1% agreement). The Passion-driven cluster maps predominantly to the k-modes Passion cluster (84.2%). The Sport & social cluster and Family-influenced cluster show more dispersion, reflecting greater internal heterogeneity in those groups.

The moderate overall ARI (0.270) is consistent with expectations when comparing algorithms that operate on different representations of the same data, one on raw binary items and the other on aggregated construct indices. The key substantive finding is that both algorithms independently identify Passion, Financial, and Family as distinct motivational axes, and both reproduce an All-round cluster. The splitting of Sport & social into two sub-groups in k-modes suggests that this profile may merit further investigation in future work.

The k-modes replication supports the substantive validity of the main solution. The four theoretically distinctive profiles (Passion-driven, Financial, Family-influenced) are reproduced by an independent algorithm. The Sport & social cluster is the least stable and should be interpreted with appropriate caution.

Appendix B

Federation Confound: Per-federation Cluster Solutions

The overall k=5 solution reveals a substantial association between cluster membership and federation (Cramér's V = 0.273, $p < .001$), raising the question of whether the five motivational profiles reflect genuine differences in referee motivation or are partly an artefact of the national composition of the sample. To address this, the clustering was re-run independently within each of the three federations using the same algorithm, the same eight construct indices, and k=5.

B.1 Method

For each federation sub-sample, k-means clustering was applied to the eight standardised construct indices with k=5 and n_init=20. The same Standard_Scaler fitted on the full dataset was applied within each sub-sample to ensure comparability of construct index values across federations. Agreement between the per-federation solutions and the overall solution was assessed using the Adjusted Rand Index (ARI). Silhouette scores are reported as a quality indicator for each sub-solution.

B.2 Results

Table B1. Country distribution.

England (n=1,811, silhouette=0.226, ARI vs overall=0.409)

Cl.	n	% fed	Defining construct
C1	544	30.0%	Sport & fitness (0.643)
C2	542	29.9%	Community (0.533)
C3	356	19.7%	Financial (0.708)
C4	349	19.3%	Family (0.523)
C5	20	1.1%	Passion (1.000)

Spain (n=2,036, silhouette=0.214, ARI vs overall=0.560)

Cl.	n	% fed	Defining construct
C1	638	31.3%	Financial (0.938)
C2	487	23.9%	Sport & fitness (0.752)
C3	446	21.9%	Passion (0.598)
C4	237	11.6%	Learning & dev (0.504)
C5	228	11.2%	Family (0.502)

Italy (n=4,740, silhouette=0.216, ARI vs overall=0.286)

Cl.	n	% fed	Defining construct
C1	1,567	33.1%	Passion (0.639)
C2	1,196	25.2%	Passion (secondary) (0.548)
C3	1,130	23.8%	Financial (0.730)
C4	763	16.1%	Learning & dev (0.510)
C5	84	1.8%	Passion (All-round) (0.988)

Note on dropout analysis: the Italian federation (n=4,740) did not collect the "no_continuation" variable that identifies inactive referees. Dropout comparisons in this appendix are therefore restricted to England (n=1,811) and Spain (n=2,036). This limitation applies to any federation-level dropout interpretation throughout the paper and is reported as a study limitation in the main text.

B.3 Cross-federation comparison

Table B2 compares the defining construct of each cluster across the three federations. The Financial cluster is recovered in all three federations, though it is considerably more pronounced in Spain (defining value 0.938) than in Italy (0.730) or England (0.708). The Family-influenced cluster is recovered in all three federations, appearing as a distinct profile in England (19.3%, n=349), Spain (11.2%, n=228), and Italy (1.8%, n=84).

The Passion-driven profile is recovered in Italy and Spain but manifests differently in England, where Community emerges as the dominant construct in one cluster (n=542), suggesting that English referees who are intrinsically motivated may express that motivation through altruistic and community-oriented language rather than personal passion. This cross-national variation is theoretically interesting and warrants further investigation.

Table B2. Defining construct per cluster across federations

	England (n=1,811)	Spain (n=2,036)	Italy (n=4,740)
C1	Sport & fitness (0.643, 30%)	Financial (0.938, 31%)	Passion (0.639, 33%)
C2	Community (0.533, 30%)	Sport & fitness (0.752, 24%)	Passion secondary (0.548, 25%)
C3	Financial (0.708, 20%)	Passion (0.598, 22%)	Financial (0.730, 24%)
C4	Family (0.523, 19%)	Learning & dev (0.504, 12%)	Learning & dev (0.510, 16%)

	England (n=1,811)	Spain (n=2,036)	Italy (n=4,740)
C5	Passion / All-round (1.000, 1%)	Family (0.502, 11%)	All-round (0.988, 2%)
ARI	0.409 (moderate)	0.560 (moderate–good)	0.286 (moderate)

Values show the defining (highest-mean) construct index for each cluster within each federation's independent solution. Matching colours indicate conceptual correspondence across federations.

B.4 Conclusion

The per-federation analysis supports the substantive validity of the overall typology in two important respects. First, the Financial cluster is recovered in all three federations with consistent magnitude, confirming that financial motivation as a distinct referee profile is not a national artefact. Second, the Family-influenced cluster is present in all three federations, including Italy, despite Italian referees being under-represented in that profile in the overall solution. This confirms that family-triggered entry is a cross-national phenomenon, not an English peculiarity.

The Community cluster appearing in England (but not Spain or Italy) and the prominence of Learning & development in Spain and Italy (but not England) suggest some cross-national differences in the motivational landscape likely reflecting the structure of referee training programmes, the role of youth football volunteering, or cultural differences in sport participation across the three countries.

The federation confound does not undermine the overall typology. The four theoretically central profiles (Financial, Passion-driven, Sport & fitness, and Family-influenced) are recovered in all three national contexts, with nationally specific variations in prominence and expression that are themselves theoretically informative.

Appendix C

Table C1. Survey Questionnaire Variables and Response Coding.

Field	Definition	Response options or coding
id	Response-only identifier	Numeric identifier
fed	Federation	EN = England; ES = Spain; IT = Italy
Top_category_level	Highest level reached	0 = not answered; 1 to 10, where 1 = lowest level
Current_category_level	Current level	0 = not answered; 1 to 10, where 1 = lowest level
fam_status_group	Family status	D = divorced; M = married; S = single; V = widow/widower; 0 = not answered
Gender_man	Gender	0 = woman; 1 = man
Born_Year	Year of birth	Year
Start_Year	Year started refereeing	Year
Start_Age	Age started refereeing	Age in years
Regional_Fed	Regional federation	Name of the regional federation
Seniority (Tenure)	Years of experience in refereeing	Number of years
Cat_by_Seniority	Current level divided by seniority (tenure)	Ratio
Seniority_interval	Tenure interval	1 to 6
Place of birth	City of birth	Name of city
RtS_interesting	Reason to start refereeing: It was appealing to me	1 = yes; 0 = no

Field	Definition	Response options or coding
RtS_others	Reason to start refereeing: Because my friend(s) or relative(s) did	1 = yes; 0 = no
RtS_money	Reason to start refereeing: Financial reward	1 = yes; 0 = no
RtS_stay	Reason to start refereeing: To remain involved with a sport after retiring from playing	1 = yes; 0 = no
RtS_family	Reason to start refereeing: Family member encouragement	1 = yes; 0 = no
RtS_mychildren	Reason to start refereeing: My son/daughter played sport and their team needed an official	1 = yes; 0 = no
RtS_giveback	Reason to start refereeing: Giving something back to the community	1 = yes; 0 = no
RtS_challenge	Reason to start refereeing: Like a personal challenge	1 = yes; 0 = no
RtS_sport	Reason to start refereeing: To practise sport or stay fit	1 = yes; 0 = no
RtS_social	Reason to start refereeing: Participating in a social activity	1 = yes; 0 = no
RtS_enthusiasm	Reason to start refereeing: Enthusiasm or passion for a sport	1 = yes; 0 = no
RtS_learn	Reason to start refereeing: Learning new skills	1 = yes; 0 = no
RtS_volunt	Reason to start refereeing: Satisfaction with achievements as a volunteer	1 = yes; 0 = no
RtS_resume	Reason to start refereeing: To improve my CV	1 = yes; 0 = no
RtS_other	Reason to start refereeing: Other	1 = yes; 0 = no
RtL_agression	Reason to leave: Suffering aggression or being threatened with aggression	1 = yes; 0 = no

Field	Definition	Response options or coding
RtL_shape	Reason to leave: Lack of physical condition or injury	1 = yes; 0 = no
RtL_dissap	Reason to leave: Disappointment at not being included in development programs	1 = yes; 0 = no
RtL_incompPers	Reason to leave: Incompatibility with personal circumstances	1 = yes; 0 = no
RtL_incompProf	Reason to leave: Incompatibility with professional circumstances	1 = yes; 0 = no
RtL_loseInt	Reason to leave: Loss of interest	1 = yes; 0 = no
RtL_lackFollow	Reason to leave: Lack of follow-up from the federation	1 = yes; 0 = no
RtL_requir	Reason to leave: Physical requirements and tests are too onerous	1 = yes; 0 = no
RtL_dissag	Reason to leave: Disagreement with the response received from the federation	1 = yes; 0 = no
RtL_remunera	Reason to leave: Insufficient remuneration	1 = yes; 0 = no
RtL_demotion	Reason to leave: Demotion	1 = yes; 0 = no
RtL_unfair	Reason to leave: Unfair promotions or evaluations	1 = yes; 0 = no
RtL_ageLimitOR other	Reason to leave: Age limit or other reason	1 = yes; 0 = no
no_continuation	No longer active in refereeing	1 = yes; 0 = no
cont_delegate	Continued as delegate	1 = yes; 0 = no
cont_mentor	Continued as mentor	1 = yes; 0 = no
cont_coachtrain	Continued as coach or trainer	1 = yes; 0 = no

Field	Definition	Response options or coding
cont_manager	Continued as manager in the federation	1 = yes; 0 = no
cont_repr	Continued as representative in the federation	1 = yes; 0 = no
cont_admin	Continued in administration in the federation	1 = yes; 0 = no
cont_mark	Continued in marketing in the federation	1 = yes; 0 = no
cont_psi	Continued as psychologist	1 = yes; 0 = no
famORfriend	Influence to start from a relative or friend	1 = yes; 0 = no
playORcoach	Influence to start because respondent previously played football or coached	1 = yes; 0 = no
ini_vocation	Initial vocation for refereeing	1 = yes; 0 = no
RtC_promo	Reason to continue: Promotion expectations	1 = yes; 0 = no
RtC_money	Reason to continue: Extra income	1 = yes; 0 = no
RtC_refstatus	Reason to continue: Social status of referees	1 = yes; 0 = no
RtC_fedtreat	Reason to continue: Differential treatment from the federation	1 = yes; 0 = no
RtC_stay	Reason to continue: Excitement to continue refereeing	1 = yes; 0 = no
RtC_relation	Reason to continue: Relationship with other referees	1 = yes; 0 = no
RtC_sport	Reason to continue: To practise sport or stay fit	1 = yes; 0 = no
RtC_social	Reason to continue: Enjoyment of the social activity that refereeing represents	1 = yes; 0 = no
RtC_other	Reason to continue: Other	1 = yes; 0 = no
Ready_Promo	Respondent considered themselves ready for promotion	1 = yes; 0 = no

Field	Definition	Response options or coding
threaten	Experienced a situation in which they felt physically threatened	0 = no; 1 = occasionally; 2 = sometimes; 3 = many times
coach_in	Attitudes of coaches regarding refereeing during matches	3 = collaborative; 2 = neutral; 1 = confrontational; 0 = hostile
coach_off	Attitudes of coaches regarding refereeing outside matches	3 = collaborative; 2 = neutral; 1 = confrontational; 0 = hostile
fans_in	Attitudes of spectators regarding refereeing during matches	3 = collaborative; 2 = neutral; 1 = confrontational; 0 = hostile
fans_off	Attitudes of spectators regarding refereeing outside matches	3 = collaborative; 2 = neutral; 1 = confrontational; 0 = hostile
reps_in	Attitudes of team representatives regarding refereeing during matches	3 = collaborative; 2 = neutral; 1 = confrontational; 0 = hostile
reps_off	Attitudes of team representatives regarding refereeing outside matches	3 = collaborative; 2 = neutral; 1 = confrontational; 0 = hostile
media_in	Attitudes of media regarding refereeing during matches	3 = collaborative; 2 = neutral; 1 = confrontational; 0 = hostile
media_off	Attitudes of media regarding refereeing outside matches	3 = collaborative; 2 = neutral; 1 = confrontational; 0 = hostile
refs_in	Attitudes of other referees regarding refereeing during matches	3 = collaborative; 2 = neutral; 1 = confrontational; 0 = hostile
refs_off	Attitudes of other referees regarding refereeing outside matches	3 = collaborative; 2 = neutral; 1 = confrontational; 0 = hostile

Field	Definition	Response options or coding
subs	Attitudes of substitutes regarding refereeing	3 = collaborative; 2 = neutral; 1 = confrontational; 0 = hostile
avoidable_drop	Whether dropout could have been avoided by the federation or was due to personal reasons	1 = yes, it could have been avoided; 0 = no, it was due to personal reasons
dev_prog	Participated in development program(s)	1 = yes; 0 = no
dev_prog_idlike	Did not participate in development program(s) but would have liked to	1 = yes; 0 = no
progr_dev	Development program(s) participated in	Name of program
dependents	Responsible for dependents	1 = yes; 0 = no
dependents#	Number of dependents	Number of dependents
academic_level	Academic level reached	0 = not answered; 1 to 6, where 1 = lowest level

RtS = reasons to start; RtC = reasons to continue; RtL = reasons to leave. Binary variables are coded 1 = yes and 0 = no unless otherwise indicated. Variable names are reported as provided in the survey dataset. Verify the exact spelling of RtL_agression, RtL_dissap, RtL_dissag, RtL_ageLimitOR other, and dependents# before final submission if harmonised naming is required.